Concept Disentangled Transformer for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification remains challenging due to high spectral dimensionality, extremely limited labeled samples, and complex spatial–spectral dependencies. Recent Transformer-based models provide strong contextual modeling, yet their high-dimensional latent representations are often weakly structured under low-label regimes, which can lead to unstable class separation and limited analyzability of the decision process. This article proposes CDFormer (CDF), a Transformer framework that introduces an explicit concept bottleneck (CD) to impose a structured intermediate representation for classification. CDF projects attention-enhanced token features into a low-dimensional concept space composed of structured latent coordinates for decision decomposition and predicts labels through a linear class-concept readout, yielding an exact logit decomposition into concept-wise contributions rather than predefined semantic attributes. To improve representation learning under scarce supervision, we train CDF using a composite objective that combines standard classification loss with orthogonality and sparsity regularization (SP) and a supervised contrastive term applied in concept space, encouraging reduced feature redundancy, selective coordinate activation, and improved intraclass compactness/interclass separation. Experiments on five benchmark HSI datasets demonstrate that CDF achieves consistent performance compared with recent state-of-the-art baselines.**

***Index Terms*—Concept learning, hyperspectral image (HSI) classification, multiloss objective, structural transparency, transformer.**

Nomenclature

$\mathcal{X} \in \mathbb{R}^{H\times W_{spa}\times B}$	HSI cube.
$H,\ W_{spa},\ B$	Height, width, and bands.
$\mathcal{P}_i \in \mathbb{R}^{P\times P\times B}$	Spatial–spectral patch.
$M = P^2$	Number of tokens.
$\mathbf{X}_0 \in \mathbb{R}^{M\times d_{\text{in}}}$	Initial token embedding.
$\mathbf{x}_j \in \mathbb{R}^{d_{\text{in}}}$	Embedded representation of token j.
$\mathbf{P} \in \mathbb{R}^{M\times d_{\text{in}}}$	Learnable positional encoding.
$\mathbf{X} \in \mathbb{R}^{M\times d_{\text{in}}}$	Input token matrix to the transformer.
d_k & h	Dimensions and number of attention heads.
$\mathbf{Q}^{(i)}, \mathbf{K}^{(i)}, \mathbf{V}^{(i)} \in \mathbb{R}^{M\times d_k}$	Query, key, and value matrices of head i.
$\mathbf{W}_Q^{(i)}, \mathbf{W}_K^{(i)}, \mathbf{W}_V^{(i)} \in \mathbb{R}^{d_{\text{in}}\times d_k}$	Attention projection matrices.
$\mathbf{W}_O \in \mathbb{R}^{hd_k\times d_{\text{in}}}$	Output projection matrix.
$\mathbf{W}_1,\ \mathbf{W}_2$	Feed-forward network weight matrices.
$\mathbf{W}_c \in \mathbb{R}^{d_h\times K}$	Concept projection matrix.
K	Number of concept dimensions.
$\mathbf{z}_{ij} \in \mathbb{R}^K$	Token-level concept vector.
$\mathbf{r}_i \in \mathbb{R}^K$	Pooled concept representation of sample i.
$\mathbf{W}_{\text{cls}} \in \mathbb{R}^{K\times C_l}$	Class-concept classifier matrix.
C_l	Number of classes.
$\hat{\mathbf{p}}_i \in \mathbb{R}^{C_l}$	Predicted class probability vector.
$y_{ic} \in \{0, 1\}$	Ground-truth class indicator.
$\mathcal{L}_{\text{cls}}$	Cross-entropy classification loss.
$\mathcal{L}_{\text{ortho}}$	Orthogonality regularization loss.
$mathcalL_{\text{sparse}}$	Sparsity regularization loss.
$\mathcal{L}_{\text{con}}$	Supervised contrastive loss.
$\mathcal{L}_{\text{total}}$	Total training objective.
$\lambda_1, \lambda_2, \lambda_3$	Loss weighting coefficients.

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I. Introduction

HYPERSPECTRAL image (HSI) classification plays a central role in remote sensing by enabling fine-grained material identification through rich spectral signatures across hundreds of contiguous bands [1]. The ability to extract meaningful spectral–spatial patterns has driven advances in environmental monitoring [2], [3], precision agriculture [4], urban infrastructure analysis [5], and mineral exploration [6]. Despite recent progress, two persistent challenges remain: 1) effective learning under limited annotated data [7]; and 2) limited structural transparency of deep neural networks, whose high-dimensional latent representations are difficult to analyze [8], [9].

Recent deep learning approaches, including convolutional neural networks (CNNs) and Transformer-based architectures,

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have significantly improved HSI classification performance [10], [11]. In particular, Transformer models employ multihead self-attention (MHSA) to capture long-range spectral–spatial dependencies [12], [13]. However, these models typically learn distributed and entangled feature representations optimized primarily for classification accuracy [14], [15]. The final prediction is commonly obtained through a direct linear readout of high-dimensional embeddings, providing limited insight into intermediate structured variables that mediate the decision process and may support representation analysis under low-label settings [16], [17]. Moreover, the lack of structured intermediate representations becomes particularly critical in high-stakes remote sensing scenarios such as disaster response and land use monitoring, where transparency, traceability, and reliability of model predictions are essential [18].

Recent studies have also emphasized uncertainty-aware learning, representation nonuniformity, and robust feature utilization for hyperspectral analysis and remote sensing perception tasks [19], [20], [21]. These developments further highlight the importance of learning compact and informative representations under challenging supervision conditions.

Concept-based learning has emerged as a promising direction for introducing structured intermediate representations into neural systems. Concept learning aims to organize internal features into lower-dimensional structured latent variables that mediate the prediction process and provide a more analyzable representation space [22], [23], [24]. However, existing concept-level approaches often face limitations: many require human-annotated concept labels [22], may suffer from entangled or redundant concept representations [23], [24], or introduce architectural complexity that limits integration with high-performance Transformer backbones [25], [26], [27]. Furthermore, concept annotations are often difficult to obtain in hyperspectral applications, where explicit semantic concepts may not be readily available. These limitations motivate the development of an annotation-free and scalable framework that embeds structured latent representations directly within modern attention-based architectures while preserving the spectral–spatial modeling strengths of Transformer encoders.

To address these challenges, this work proposes CDFormer (CDF), a Transformer-based framework that integrates an explicit concept bottleneck (CD) within the encoder architecture. Instead of predicting classes directly from high-dimensional token embeddings, CDF projects attention-enhanced features into a lower-dimensional concept space composed of structured latent coordinates and performs classification through a linear class-concept mapping. This design yields an exact additive decomposition of class logits in terms of concept activations, enabling structural transparency of the decision function through an analyzable intermediate representation.

To improve representation quality under limited supervision, CDF employs a composite objective consisting of classification loss, orthogonality regularization (OR), sparsity regularization (SP), and supervised contrastive learning (CL). Orthogonality reduces redundancy among concept directions, sparsity promotes selective activation, and CL encourages intraclass compactness and interclass separation. Collectively, these mechanisms help organize the learned representation space while improving stability and generalization in low-label HSI classification settings. Importantly, this process does not require manually annotated concept labels.

Built upon a Transformer encoder with MHSA and learnable positional encodings, CDF preserves the spectral–spatial modeling strengths of attention-based architectures while introducing structured latent representations for decision decomposition. Extensive experiments on five benchmark HSI datasets demonstrate consistent performance under low-label settings.

II. Related Works

HSI classification has evolved significantly with advances in deep learning, transitioning from traditional pixel-wise methods to spectral–spatial architectures capable of modeling high-dimensional dependencies. Recent progress has largely focused on improving representation capacity through CNNs, Transformer-based architectures, and hybrid attention mechanisms. Despite improvements in predictive performance, challenges remain in terms of representation structure and robustness under limited supervision. In this section, we review relevant developments in spectral–spatial modeling, attention-based architectures, and concept-level learning, and clarify how the proposed CDF is positioned within this landscape.

A. Spectral–Spatial and Attention-Based Modeling in HSI

Early deep learning approaches leveraged 2-D and 3-D CNNs to jointly capture spectral and spatial information. While effective, 3-D convolutions often introduce high computational complexity and limited receptive field flexibility. To address long-range dependencies more effectively, Transformer-based models have recently been introduced for HSI classification.

SpectralFormer [28] modeled spectral sequences using MHSA mechanisms, significantly improving the modeling of global spectral correlations. Similarly, Li et al. [29] proposed a cascaded Transformer framework for fusion-aware computational HSI. Although these approaches demonstrate strong performance, their learned representations remain distributed across high-dimensional latent spaces without explicit structural constraints. Consequently, interpretability typically relies on indirect inspection of attention weights rather than structured intermediate representations. Furthermore, graph-based Transformer approaches such as GraphGST [30] and GGCN [31] incorporate topology-aware inductive biases by modifying token interactions or constructing similarity graphs. These methods improve relational modeling among spectral–spatial samples by explicitly exploiting graph structures and neighborhood dependencies. However, their primary objective is to enhance contextual information propagation and graph-aware feature aggregation rather than to learn structured intermediate representations for decision decomposition.

In contrast to graph-based approaches, which primarily focus on relational modeling, another line of research has incorporated explainability through post hoc analysis. For

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instance, De Lucia et al. [32] introduced spectral-cumulative Grad-CAM for 3-D CNNs to provide visual explanations in edge computing scenarios. Atik and Atik [33] employed SHAP and permutation feature importance (PFI) for spectral band selection combined with deep reinforcement learning. Temenos et al. [34] integrated SHAP into compact CNN architectures for land use and land cover (LCLU) classification. Similarly, Ebrahimi and Kumar [35] proposed interclass-Grad-CAM and spectral-Grad-CAM for semantic segmentation tasks such as crop and LCLU mapping [36].

While these post hoc techniques provide valuable visualization tools, they operate externally to the core model architecture. Their explanations are derived from gradient sensitivity or feature perturbation analyses, rather than from explicitly structured intermediate representations embedded within the model. As a result, they quantify local sensitivity rather than providing an algebraic decomposition of decision variables.

B. Modeling Spectral Variability and Structured Representations

Beyond attention-based architectures, several works have addressed spectral variability and representation robustness. Hong et al. [37] proposed a nonconvex ALMM model to better capture spectral variability in high-dimensional data. Kakogeorgiou and Karantzalos [38] explored interpretability techniques for multilabel classification. Shiet al. [39] introduced BIT-DNN, integrating biophysical and biochemical attributes to enhance vegetation classification interpretability.

Although these methods incorporate domain knowledge, auxiliary attributes, and representation-guiding mechanisms, they often require additional modeling complexity or task-specific constraints. Moreover, many approaches focus on visualization, attribute integration, or spectral variability modeling without explicitly constraining the geometry of the learned latent representation space.

Recently, large-scale vision foundation models and large multimodal models have demonstrated remarkable success across a wide range of remote sensing and computer vision tasks by leveraging large-scale pretraining and transferable representations [27], [40], [41], [42], [43]. Several studies have explored adapting foundation-model paradigms to HSI analysis through self-supervised learning [44], spectral–spatial pretraining [45], [46], and cross-modal representation alignment [47]. These approaches generally benefit from extensive pretraining data and substantial computational resources, enabling improved generalization when sufficient data are available. However, HSI classification frequently operates in highly label-scarce environments, where acquiring large-scale annotated datasets remains challenging. Moreover, the high spectral dimensionality, domain heterogeneity, and computational requirements of large models may limit their practical deployment in resource-constrained remote sensing applications. Consequently, lightweight and label-efficient architectures remain highly relevant for HSI classification.

While foundation-model approaches emphasize large-scale pretraining and transferable representations, an alternative direction is to improve representation organization and decision structure directly within task-specific models, particularly under limited-label supervision.

C. Concept-Based Learning Frameworks

Concept-based learning has emerged as a promising direction for introducing structured intermediate representations into neural architectures. Rather than operating solely on high-dimensional latent embeddings, concept-based approaches organize internal representations into lower-dimensional variables that provide a more analyzable decision structure. CD models [22] explicitly predict predefined semantic concepts before final classification. Extensions such as [23], [24] investigate structured concept learning and geometry-aware representation organization.

However, existing concept-learning frameworks typically rely on manually annotated concept labels, which are difficult to obtain in hyperspectral settings. Furthermore, without appropriate geometric constraints, learned concept dimensions may remain entangled, limiting representation quality and robustness. Some works [25], [26] incorporate structured reasoning or graph-based mechanisms, yet these approaches often increase architectural complexity or introduce additional supervision requirements.

Collectively, these observations highlight three recurring gaps in existing approaches. First, high-capacity spectral–spatial models provide strong predictive performance but generally lack explicitly structured intermediate representations. Second, post hoc explainability techniques analyze trained models without influencing the representation learning process itself. Third, concept-based frameworks improve representation organization but frequently rely on concept annotations or introduce additional modeling complexity. Motivated by these limitations, CDF integrates a structured CD directly within a Transformer encoder and learns latent concept representations solely from class supervision through OR, sparsity constraints, and supervised CL. This design enables annotation-free structured representation learning while preserving the spectral–spatial modeling strengths of attention-based architectures.

D. Positioning of CDF

CDF is designed to address the limitations identified above. Unlike post hoc explainability approaches [32], [33], [34], [35], the proposed framework incorporates concept representations into the prediction process itself through a class-concept mapping. Unlike CD models that rely on manually annotated concepts [22], CDF learns structured latent concept representations directly from class supervision without requiring concept annotations. Furthermore, in contrast to graph-based and attribute-driven approaches [25], [26], [39], CDF introduces representation-level structure through orthogonality, sparsity, and contrastive regularization while preserving the spectral–spatial modeling strengths of Transformer architectures. Consequently, the proposed framework seeks to improve representation organization, structural transparency, and generalization under limited-label HSI classification settings. Table I

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TABLE I

COMPARISON OF CONVENTIONAL END-TO-END HSI CLASSIFIERS, CONCEPT-LEVEL LEARNING FRAMEWORKS, AND THE PROPOSED CDF

Aspect	Pixel/End-to-End Models	Concept-Level Models	CDFormer
Representation	High-dimensional latent embedding	Explicit intermediate concept variables	Explicit Transformer-integrated concept bottleneck
Classifier	Linear/MLP readout of embedding	Class prediction via concept layer	class-concept mapping with additive logit
Transparency	Post-hoc attribution	Predefined semantic concepts	Intrinsic algebraic class-concept decomposition
Concept Supervision	Class labels only	Often requires concept annotations	Class labels only (annotation-free structured concept learning)
Integration	CNN/Transformer backbone	Separate concept module	Concept projection embedded within Transformer
Objective Structure	Classification loss	Classification + concept constraints	Classification + orthogonality + sparsity + contrastive regularization
Modeling Scope	Performance-oriented latent model	Structured representation modeling	Structured representation learning

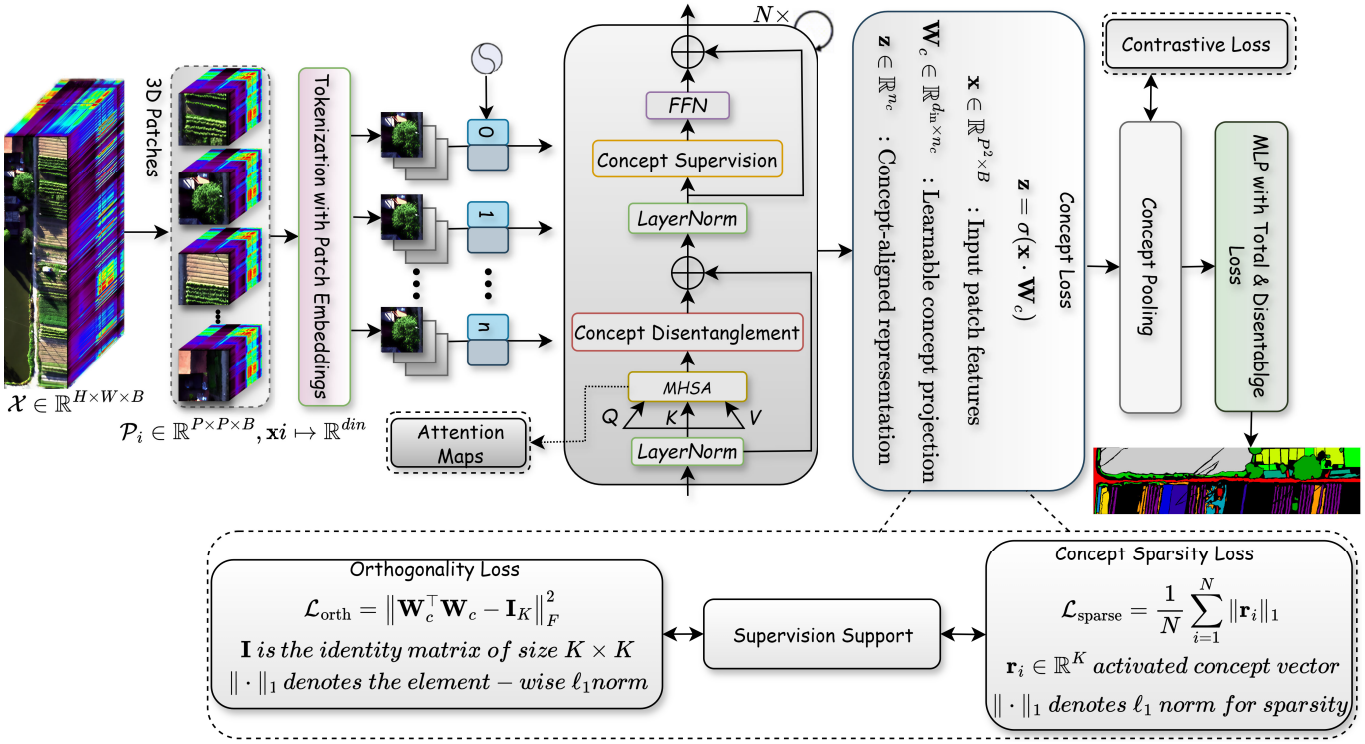


Fig. 1. Overview of the proposed CDF architecture for HSI classification. The input spectral–spatial cube is first processed by a patch embedding block, which jointly encodes spectral signatures and spatial textures into a sequence of patch tokens. These tokens are then passed to the concept transformer, which employs MHSA and feed-forward layers to capture long-range dependencies and global contextual interactions. Concept projection together with orthogonality, sparsity, and contrastive regularization further structure the learned representations.

summarizes the key differences between conventional end-to-end HSI classifiers, existing concept-level approaches, and the proposed CDF framework.

III. METHODOLOGY

The proposed CDF framework integrates spatial–spectral tokenization, Transformer-based contextual modeling, concept projection, and multiobjective optimization to enable structured latent representation learning and decision decomposition for HSI classification. Within CDF, concepts are modeled as structured latent coordinates that organize intermediate representations into an analyzable concept space rather than predefined remote sensing semantic attributes. The proposed concept space is learned directly from class supervision and regularization objectives without requiring concept annotations. For clarity and to avoid ambiguity in symbol usage, Nomenclature summarizes the principal notations and definitions used throughout Section III. Fig. 1 and Algorithm 1 summarize the end-to-end pipeline.

A. Patch Extraction and Feature Embedding

An HSI is represented as a 3-D tensor $\mathcal{X} \in \mathbb{R}^{H \times W_{spa} \times B}$, where H and W_{spa} denote the spatial height and width, respectively, and B is the number of spectral bands. To preserve both spatial and spectral context, the image is divided into overlapping patches using a sliding window of size $P \times P$ and stride s. Each resulting patch is denoted by $\mathcal{P}_i \in \mathbb{R}^{P \times P \times B}$ and captures local spectral–spatial structures [48], [49]. For each patch, spectral–spatial observations are converted into a sequence of token representations and projected into a shared embedding space through a learnable linear projection. The resulting token sequence is represented as

$$\mathbf{X}_0 = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_M]^\top \in \mathbb{R}^{M \times d_{\text{in}}} \quad (1)$$

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Algorithm 1 CDFormer

Initialize learnable parameters Θ.
while not converged **do**
  mini-batch of labeled centers $\{(u_i, v_i, y_i)\}_{i=1}^{N_b} \subset \Omega$.
  for $i = 1$ **to** N_b **do**
    Extract spatial-spectral patch.
    Tokenize $\mathcal{P}_i$ into M sub-patches with stride s.
    Compute patch tokens via Patch Embedding.
    Add positional encoding: $\mathbf{X}_i \leftarrow \mathbf{X}_i + \mathbf{P}$.
    for $\ell = 1$ **to** L **do**
      Token normalization: $\mathbf{H}_i^{(\ell)} \leftarrow \text{LayerNorm}(\mathbf{X}_i)$
      Self-attention: $\mathbf{A}_i^{(\ell)} \leftarrow \text{MHSA}_\ell(\mathbf{H}_i^{(\ell)})$
      Concept projection: $\mathbf{Z}_i^{(\ell)} \leftarrow \mathbf{A}_i^{(\ell)}\mathbf{W}_c \in \mathbb{R}^{M\times K}$
      Concept activation: $\mathbf{R}_i^{(\ell)} \leftarrow \sigma(\mathbf{Z}_i^{(\ell)}) \in \mathbb{R}^{M\times K}$
      Residual fusion: $\mathbf{X}_i \leftarrow \mathbf{H}_i^{(\ell)} + \mathbf{R}_i^{(\ell)}\mathbf{P}_c^\top$
      Post-fusion normalization.
      FFN block: $\mathbf{F}_i^{(\ell)} \leftarrow \text{FFN}_\ell(\mathbf{X}_i)$
      Residual update: $\mathbf{X}_i \leftarrow \mathbf{X}_i + \mathbf{F}_i^{(\ell)}$
    end for
    Concept pooling: $\bar{\mathbf{r}}_i \leftarrow \frac{1}{M}\sum_{j=1}^{M} \mathbf{R}_{i,j}^{(L)} \in \mathbb{R}^K$.
    Class logits: $\hat{\mathbf{p}}_i \leftarrow \text{softmax}(\mathbf{W}_{\text{cls}}^\top \bar{\mathbf{r}}_i) \in \mathbb{R}^{C_l}$.
  end for
  Classification loss: $\mathcal{L}_{\text{cls}} \leftarrow \frac{1}{N_b}\sum_{i=1}^{N_b} \text{CE}(\mathbf{y}_i, \hat{\mathbf{p}}_i)$.
  Orthogonality: $\mathcal{L}_{\text{ortho}} \leftarrow \|\mathbf{W}_c^\top \mathbf{W}_c - \mathbf{I}_K\|_F^2$.
  Sparsity: $\mathcal{L}_{\text{sparse}} \leftarrow \frac{1}{N_b}\sum_{i=1}^{N_b} \|\bar{\mathbf{r}}_i\|_1$.
  Supervised contrastive loss: $\mathcal{L}_{\text{con}} \leftarrow 0$.
  for each anchor $i \in \{1, \ldots, N_b\}$ **do**
    Select a positive index j^+ such that $y_{j^+} = y_i$, $j^+ \neq i$.
    $\mathcal{L}_{\text{con}} \leftarrow \mathcal{L}_{\text{con}} - \log \frac{\exp(\text{sim}(\bar{\mathbf{r}}_i, \bar{\mathbf{r}}_{j^+})/\tau)}{\sum_{k\in\{1,\ldots,N_b\}\setminus\{i\}} \exp(\text{sim}(\bar{\mathbf{r}}_i, \bar{\mathbf{r}}_k)/\tau)}$.
  end for
  $\mathcal{L}_{\text{con}} \leftarrow \frac{1}{N_b}\mathcal{L}_{\text{con}}$, with $\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}$.
  Total objective:
  $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{ortho}} + \lambda_2 \mathcal{L}_{\text{sparse}} + \lambda_3 \mathcal{L}_{\text{con}}$.
  Update: $\Theta \leftarrow Opt.step(\nabla_\Theta \mathcal{L}_{\text{total}})$.
end while
Return trained parameters Θ.

where $\mathbf{x}_j \in \mathbb{R}^{d_{\text{in}}}$ denotes the embedded representation of the jth token, $M = P^2$ is the number of spatial tokens extracted from the input patch, and d_{in} denotes the embedding dimension. This tokenization strategy preserves local spectral–spatial structure while providing a compact representation suitable for Transformer-based contextual modeling. The resulting token matrix $\mathbf{X}_0$ serves as the input to the Transformer encoder, where contextualized spectral–spatial representations are subsequently learned and projected into the concept space.

B. Transformer Components

Having defined the input tokenization, we now describe the Transformer encoder used to model long-range spatial–spectral dependencies among tokens.

1) Multihead Self-Attention: Given a token matrix $\mathbf{X}_0 \in \mathbb{R}^{M \times d_{\text{in}}}$, MHSA splits features into h heads. For each head i, attention is computed as

$$\text{Attention}^{(i)}(\mathbf{X}_0) = \text{softmax}\left(\frac{\mathbf{Q}^{(i)}\mathbf{K}^{(i)\top}}{\sqrt{d_k}}\right)\mathbf{V}^{(i)} \tag{2}$$

where $\mathbf{Q}^{(i)} = \mathbf{X}\mathbf{W}_Q^{(i)}$, $\mathbf{K}^{(i)} = \mathbf{X}\mathbf{W}_K^{(i)}$, $\mathbf{V}^{(i)} = \mathbf{X}\mathbf{W}_V^{(i)}$, and $\mathbf{W}_Q^{(i)}, \mathbf{W}_K^{(i)}, \mathbf{W}_V^{(i)} \in \mathbb{R}^{d_{\text{in}} \times d_k}$ are learned projections. Outputs from all heads are concatenated and projected

$$\text{MHSA}(\mathbf{X}_0) = \text{Concat}\left(\text{Attention}^{(1)}, \ldots, \text{Attention}^{(h)}\right)\mathbf{W}_O$$

where $\mathbf{W}_O \in \mathbb{R}^{hd_k \times d_{\text{in}}}$.

2) FFN: The FFN applies two linear transformations with a nonlinearity in between

$$\mathbf{y} = \text{ReLU}\big(\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1\big)\mathbf{W}_2 + \mathbf{b}_2 \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_{\text{in}} \times d_{ff}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{ff} \times d_{\text{in}}}$, and $\mathbf{b}_1, \mathbf{b}_2$ are biases. The FFN is applied independently to each token position.

3) Positional Encoding: Transformers lack inherent spatial ordering. To retain spatial structure, learnable positional encodings $\mathbf{P} \in \mathbb{R}^{M \times d_{\text{in}}}$ are added

$$\mathbf{X}_{\text{pos}} = \mathbf{X} + \mathbf{P}.$$

4) Transformer Encoder Layer: Each encoder layer combines MHSA and FFN with residual connections and layer normalization

$$\mathbf{Z}_1 = \text{LayerNorm}\big(\mathbf{X}_{\text{pos}} + \text{MHSA}\big(\mathbf{X}_{\text{pos}}\big)\big) \tag{4}$$

$$\mathbf{Z}_2 = \text{LayerNorm}\big(\mathbf{Z}_1 + \text{FFN}\big(\mathbf{Z}_1\big)\big) \tag{5}$$

where $\mathbf{Z}_2$ is passed to the next encoder layer. After the Transformer encoder extracts contextualized spectral–spatial representations, the resulting features are projected into the concept space described in Section III-C.

C. Loss Functions and Optimization

After defining the concept space, we now describe the learning objective that jointly optimizes classification performance and representation structure. Model training is guided by a composite loss that balances classification accuracy with constraints promoting reduced redundancy, selective concept activation, and improved class separability. These regularization mechanisms are designed to encourage compact and discriminative structured latent representations, which are particularly beneficial under limited-supervision HSI classification settings.

1) Concept-Space Classification Loss: Classification supervision is applied directly in the concept space to encourage discriminative, structured latent representations for decision decomposition. Let $\mathbf{r}_i \in \mathbb{R}^K$ denote the pooled concept representation associated with sample i. The cross-entropy (CE) classification loss is defined as

$$\mathcal{L}_{\text{cls}} = -\sum_{i=1}^{N}\sum_{c=1}^{C_l} y_{ic} \log\left(\frac{\exp\left(\mathbf{w}_c^\top \mathbf{r}_i\right)}{\sum_{j=1}^{C_l}\exp\left(\mathbf{w}_j^\top \mathbf{r}_i\right)}\right) \tag{6}$$

where C_l is the number of classes, $y_{ic} \in \{0, 1\}$ indicates whether sample i belongs to class c, and $\mathbf{w}_c$ is the classifier weight vector associated with class c. The classifier weight

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matrix is denoted by $\mathbf{W}_{\text{cls}} = [\mathbf{w}_1, \ldots, \mathbf{w}_{C_l}] \in \mathbb{R}^{K \times C_l}$. For token-level concept vectors $\mathbf{z}_{ij} \in \mathbb{R}^K$ at token position j for sample i, the pooled concept representation is defined as

$$\mathbf{r}_i = \frac{1}{M} \sum_{j=1}^{M} \mathbf{z}_{ij} \tag{7}$$

where the pooled representation is projected into the class space and passed through a softmax layer

$$\hat{\mathbf{p}}_i = \text{softmax}\left(\mathbf{W}_{\text{cls}}^{\top} \mathbf{r}_i\right). \tag{8}$$

This formulation expresses class predictions directly as a function of concept activations, yielding an explicit class-concept decomposition of the decision function rather than relying solely on a direct readout from high-dimensional latent embeddings. The corresponding dataset-level averaged CE loss is

$$\text{CE}(\mathbf{y}, \mathbf{p}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C_l} y_{ic} \log \hat{p}_{ic} \tag{9}$$

where $\mathbf{y}_i \in \mathbb{R}^{C_l}$ denotes the one-hot label vector.

2) Orthogonality and Sparsity Losses: To promote structured latent representations with reduced redundancy, CDF incorporates two regularization terms: orthogonality loss $\mathcal{L}_{\text{ortho}}$ and sparsity loss $\mathcal{L}_{\text{sparse}}$. Orthogonality encourages reduced redundancy among concept directions, whereas sparsity promotes selective activation patterns within the learned concept space. Together, these regularizers impose geometric structure on the intermediate representation and help mitigate redundancy under limited supervision. The orthogonality loss is

$$\mathcal{L}_{\text{ortho}} = \left\| \mathbf{W}_c^{\top} \mathbf{W}_c - \mathbf{I}_K \right\|_F^2 \tag{10}$$

where $\mathbf{I}_K$ is the K-dimensional identity matrix and $\|\cdot\|_F$ denotes the Frobenius norm. This regularizer encourages the columns of $\mathbf{W}_c$ to capture complementary directions in the concept space, thereby reducing linear redundancy among learned representations. The sparsity loss is defined as the average ℓ_1 norm of the activated concept vectors

$$\mathcal{L}_{\text{sparse}} = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{r}_i\|_1 \tag{11}$$

where $\|\cdot\|_1$ denotes the ℓ_1 norm. By encouraging only a subset of concept dimensions to be strongly activated for a given sample, sparsity promotes compact concept representations and reduces unnecessary feature interactions. Although orthogonality and sparsity do not imply semantic disentanglement or explicit physical correspondence of individual concepts, they encourage a more organized representation geometry with reduced redundancy and selective activation patterns. Such representation structuring facilitates decision decomposition and may improve stability and generalization under limited-label training scenarios.

3) Contrastive Loss: To further improve intraclass compactness and interclass separability within the learned concept representations, CDF incorporates a supervised contrastive loss $\mathcal{L}_{\text{con}}$. By encouraging samples from the same class to occupy nearby regions of the concept space while separating samples from different classes, this objective helps improve representation quality under limited-label supervision

$$\mathcal{L}_{\text{con}} = -\sum_{i \in \mathcal{B}} \log \frac{\exp\left(\text{sim}\left(\mathbf{r}_i, \mathbf{r}_j^{+}\right)/\tau\right)}{\sum_{k \in \mathcal{B} \setminus \{i\}} \exp\left(\text{sim}\left(\mathbf{r}_i, \mathbf{r}_k\right)/\tau\right)} \tag{12}$$

where $\mathbf{r}_i \in \mathbb{R}^K$ denotes the anchor concept representation, $\mathbf{r}_j^{+}$ is a positive sample belonging to the same class, $\mathbf{r}_k$ represents negative samples within the mini-batch, $\tau > 0$ is a temperature parameter controlling the concentration of the contrastive distribution, and $\mathcal{B}$ denotes the mini-batch index set. The similarity function is defined as cosine similarity

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^{\top} \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}. \tag{13}$$

All concept representations are ℓ_2-normalized prior to similarity computation, ensuring that similarity is determined by angular proximity rather than vector magnitude. Consequently, the contrastive objective promotes class-consistent clusters and larger interclass margins in the learned concept space, complementing the orthogonality and sparsity constraints used to structure the latent representation geometry. Such class-aware representation organization can improve discrimination and generalization when only limited labeled samples are available.

4) Total Objective Function: The complete objective combines classification supervision with complementary regularization terms that promote reduced redundancy, selective activation, and improved class separability. By jointly optimizing classification accuracy and representation geometry, the objective encourages stable and discriminative concept representations under limited-label learning conditions

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{ortho}} + \lambda_2 \mathcal{L}_{\text{sparse}} + \lambda_3 \mathcal{L}_{\text{con}} \tag{14}$$

where λ_1, λ_2, and λ_3 are positive weighting coefficients that balance orthogonality, sparsity, and contrastive regularization, respectively. The weighting coefficients control the relative contribution of each regularization component and therefore influence the geometry of the learned representation space. Collectively, the proposed objective provides a unified optimization framework that balances classification performance with representation structuring through orthogonality, sparsity, and contrastive regularization.

Theorem 1: (Capacity Control Through Concept Projection):

Let $\mathcal{F}$ denote the class of functions

$$f(\mathbf{x}) = \mathbf{W}_{\text{cls}}^{\top} \phi\left(\mathbf{W}_c^{\top} E_{\Theta}(\mathbf{x})\right) \tag{15}$$

where $\mathbf{x}$ satisfies $\|\mathbf{x}\|_2 \le R$, the encoder E_{Θ} is L_E-Lipschitz, i.e.,

$$\left\| E_{\Theta}(\mathbf{x}) - E_{\Theta}\left(\mathbf{x}'\right) \right\|_2 \le L_E \left\| \mathbf{x} - \mathbf{x}' \right\|_2 \tag{16}$$

the concept projection matrix satisfies $\|\mathbf{W}_c\|_2 \le B_c$, the classifier matrix satisfies $\|\mathbf{W}_{\text{cls}}\|_2 \le B_{\text{cls}}$, and the activation function ϕ is L_{ϕ}-Lipschitz. Then the empirical Rademacher complexity satisfies

$$\mathfrak{R}_N(\mathcal{F}) \le \frac{B_{\text{cls}} L_{\phi} B_c L_E R}{\sqrt{N}}. \tag{17}$$

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Consequently, with probability at least $1-\delta$, the generalization gap satisfies

$$\mathcal{O}\left(\frac{B_{\text{cls}}L_\phi B_c L_E R}{\sqrt{N}}\right) + \mathcal{O}\left(\sqrt{\frac{\log(1/\delta)}{N}}\right). \tag{18}$$

Proof: The empirical Rademacher complexity is defined as

$$\mathfrak{R}_N(\mathcal{F}) = \frac{1}{N}\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}}\sum_{i=1}^{N}\sigma_i f(\mathbf{x}_i)\right] \tag{19}$$

where $\sigma_i \in \{-1, +1\}$ are independent Rademacher random variables. Let $\mathbf{h} = E_\Theta(\mathbf{x})$ denote the encoder output. Since E_Θ is L_E-Lipschitz and $\|\mathbf{x}\|_2 \le R$, it follows that

$$\|\mathbf{h}\|_2 \le L_E R. \tag{20}$$

Consider the linear concept projection

$$g(\mathbf{h}) = \mathbf{W}_c^\top \mathbf{h}. \tag{21}$$

Using standard bounds for linear hypothesis classes with bounded spectral norm

$$\mathfrak{R}_N(g) \le \frac{B_c L_E R}{\sqrt{N}}. \tag{22}$$

Since ϕ is L_ϕ-Lipschitz, Talagrand's contraction lemma [50] implies

$$\mathfrak{R}_N(\phi\circ g) \le L_\phi \mathfrak{R}_N(g) \le \frac{L_\phi B_c L_E R}{\sqrt{N}}. \tag{23}$$

Finally, composing with the classifier matrix $\mathbf{W}_{\text{cls}}$ yields

$$\mathfrak{R}_N(\mathcal{F}) \le \frac{B_{\text{cls}}L_\phi B_c L_E R}{\sqrt{N}} \tag{24}$$

which proves the stated result. The above bound highlights the role of the CD as an intermediate capacity-control mechanism between the Transformer encoder and the classifier. In contrast to standard Transformer classifiers that perform prediction directly from high-dimensional latent embeddings, CDF introduces a lower-dimensional concept projection whose effective complexity is governed by the projection matrix $\mathbf{W}_c$. Consequently, controlling the norm and structure of $\mathbf{W}_c$ restricts the capacity of the downstream decision function. OR reduces redundancy among concept directions, sparsity promotes selective activation, and CL encourages compact class-specific clusters.

D. Structural Transparency and Robustness

CDF is designed with structural transparency (concept-decomposable prediction) and robustness as core principles, enabling inspection of class-concept contributions and analysis under perturbations or domain shifts.

1) Structural Transparency via Class-Concept Mapping: A key feature of CDF is the projection of high-dimensional token features into a structured concept space composed of learned latent coordinates. Each input is encoded into a concept vector $\mathbf{r}_i \in \mathbb{R}^K$, and final classification is performed through

$$\hat{\mathbf{p}}_i = \text{softmax}\left(\mathbf{W}_{\text{cls}}^\top \mathbf{r}_i\right).$$

This linear form yields an explicit class-concept decomposition of the decision function. For class c, the column $\mathbf{W}_{\text{cls}}[:,c]$ quantifies how each concept dimension contributes to the corresponding logit within the learned representation space.

2) Concept Selectivity: CDF exhibits concept selectivity, where subsets of concept dimensions activate consistently for particular classes while remaining suppressed for others. This behavior is encouraged by the sparsity and orthogonality constraints, which promote selective and less redundant concept activations. Visualizing concept activation maps $\mathbf{r}_i[k]$ over spatial locations can reveal spatially coherent activation patterns associated with the learned representation geometry. Such activation patterns provide insight into how different regions of the learned representation space contribute to class discrimination and decision decomposition. Additionally, correlating $\mathbf{r}_i$ with $\mathbf{W}_{\text{cls}}$ highlights which concept dimensions contribute most strongly to a given class decision.

3) Robustness to Noise and Domain Shifts: The CD provides a structured intermediate representation through which the sensitivity of concept activations to input perturbations can be analyzed. The following result characterizes how perturbations in the input propagate to the learned concept representations under standard Lipschitz assumptions.

Theorem 2: (Robustness of Concept Activations to Spectral Perturbations):

Let $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ be an input and let $\boldsymbol{\delta}$ be a perturbation such that $\|\boldsymbol{\delta}\|_2 \le \epsilon$. Assume the encoder E_Θ is L_E-Lipschitz, i.e., $\|E_\Theta(\mathbf{x}) - E_\Theta(\mathbf{x}')\|_2 \le L_E\|\mathbf{x}-\mathbf{x}'\|_2$, and the activation ϕ is L_ϕ-Lipschitz. Define the concept representation

$$\mathbf{r}(\mathbf{x}) = \phi\left(\mathbf{W}_c^\top E_\Theta(\mathbf{x})\right), \quad \mathbf{W}_c \in \mathbb{R}^{d_h\times K}$$

where d_h is the encoder feature dimension. Then the concept perturbation satisfies

$$\left\|\mathbf{r}(\mathbf{x}+\boldsymbol{\delta}) - \mathbf{r}(\mathbf{x})\right\|_2 \le L_\phi \left\|\mathbf{W}_c\right\|_2 L_E\,\epsilon = L_\phi\,\sigma_{\max}(\mathbf{W}_c)\,L_E\,\epsilon.$$

Proof: Let $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$. By Lipschitz continuity of ϕ

$$\begin{aligned}\left\|\mathbf{r}(\mathbf{x}') - \mathbf{r}(\mathbf{x})\right\|_2 &= \left\|\phi\left(\mathbf{W}_c^\top E_\Theta(\mathbf{x}')\right) - \phi\left(\mathbf{W}_c^\top E_\Theta(\mathbf{x})\right)\right\|_2 \\ &\le L_\phi\left\|\mathbf{W}_c^\top\left(E_\Theta(\mathbf{x}') - E_\Theta(\mathbf{x})\right)\right\|_2 \\ &\le L_\phi\left\|\mathbf{W}_c^\top\right\|_2\left\|E_\Theta(\mathbf{x}') - E_\Theta(\mathbf{x})\right\|_2 \\ &\le L_\phi\left\|\mathbf{W}_c\right\|_2 L_E\left\|\mathbf{x}'-\mathbf{x}\right\|_2 \\ &\le L_\phi\,\sigma_{\max}(\mathbf{W}_c)\,L_E\,\epsilon.\end{aligned}$$

□

The theorem shows that perturbations in the input induce bounded changes in the concept representation, with the magnitude of the perturbation controlled by the encoder Lipschitz constant, the activation Lipschitz constant, and the spectral norm of the concept projection matrix. Consequently, the CD provides a principled mechanism for analyzing representation stability under bounded spectral perturbations. While the theorem does not by itself guarantee improved robustness, it highlights how the structure and norm of the concept projection influence the sensitivity of the learned representation to input variations.

4) Decision Decomposition Through Concept Attribution: Since each class prediction is a linear combination of concept activations, CDF supports concept attribution through the

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TABLE II
SUMMARY OF THE BENCHMARK DATASETS

Name	Classes	Spatial Res.	Spectral Chan.	Platform / Acquisition	Train Ratio
HH	22	940 × 475 (0.043m)	270	DJI Matrice 600 Pro UAV	2%
HC	16	1217 × 303 (0.109m)	274	Leica Aibot X6 UAV	2%
TD	18	1740 × 860 (0.15m)	176	Gaiasky mini2-VN	2%
PG	10	1230 × 1000 (0.01m)	176	Gaiasky mini2-VN	2%
QG	6	880 × 1360 (0.15m)	176	Gaiasky mini2-VN	2%

class-concept mapping. For a predicted class c^*, the contribution of concept dimension k is

$$\alpha_{ik} = \mathbf{W}_{\text{cls}}\left[k, c^*\right] \cdot \mathbf{r}_i\left[k\right] \tag{25}$$

where α_{ik} quantifies how strongly the kth concept dimension contributes to the decision for sample i. Ranking concept dimensions by $|\alpha_{ik}|$ identifies the dominant latent coordinates contributing to the prediction. These attributions quantify contributions within the learned representation space and should not be interpreted as direct evidence of semantic or physical factors.

5) Implications: The above result establishes that the concept representation is Lipschitz-continuous [51], [52] with respect to bounded input perturbations. The sensitivity of the concept representation is governed by the encoder Lipschitz constant, the activation Lipschitz constant, and the spectral norm of the concept projection matrix. Consequently, the CD provides a principled mechanism for analyzing how perturbations propagate from encoder features to concept activations. From a representation-learning perspective, controlling the norm and structure of the projection matrix can promote more stable latent representations under bounded perturbations. Such stability may be beneficial in low-label HSI classification, where reducing sensitivity to noise and spurious variations can contribute to improved generalization. Therefore, $\mathbf{W}_c$ may be further regularized (e.g., via spectral norm penalties) to enhance representation stability.

IV. EXPERIMENTAL DATASETS AND SETTINGS

To comprehensively evaluate the proposed method, experiments were conducted on five widely used HSI datasets: WHU-Hi-HanChuan (HC), WHU-Hi-HongHu (HH), Tangdaowan (TD), Pingan (PG), and Qingyun (QG). These datasets represent diverse environments, including agricultural, urban, suburban, and coastal landscapes, with varying degrees of spectral and spatial complexity. Specifically, HC and HH capture urban/suburban scenes acquired by UAV platforms, while TD, PG, and QG contain challenging coastal terrains with high interclass similarity. Table II summarizes the main dataset statistics.

To simulate low-resource HSI classification, we use only 2% of labeled pixels for training, 2% for validation, and the remaining 96% for testing. We use stratified sampling to preserve class proportions. For each labeled pixel, a spatial–spectral patch of size 14 × 14 is extracted and used as the model input. Patches near image boundaries are handled via mirror padding, and spectral bands are normalized per band using training-set statistics. All methods are trained for 50 epochs using identical train/val/test partitions to ensure fair comparison. We use a Transformer backbone with four encoder layers and four attention heads to balance representational capacity and computational cost under limited supervision. All runtime measurements reported were obtained under identical hardware and software conditions without applying model compression techniques such as quantization, pruning, knowledge distillation, TensorRT acceleration, or other deployment-specific optimizations. Consequently, differences in inference time across datasets primarily reflect variations in spatial image size, spectral dimensionality, and the number of testing samples rather than deployment-level acceleration techniques.

We report overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). OA measures the percentage of correctly classified test samples; AA is the mean of per-class accuracies; and κ quantifies agreement beyond chance, which is informative under class imbalance. Unless otherwise stated, reported results correspond to the mean performance obtained from five independent runs with different random seeds.

TABLE III
REPRESENTATIVE HIGH-PERFORMING HYPERPARAMETER

Config.	Batch Size	Embed Dim.	LR	λ_1	λ_2	λ_3
Top 1	128	256	1e-5	0.01	0.01	0.01
Top 2	64	256	1e-5	1.0	1.0	0.01
Top 3	32	56	1e-3	0.01	1.0	1.0

TABLE IV
SENSITIVITY ANALYSIS OF THE OR COEFFICIENT λ_1 ON THE HC DATASET

λ_1	OA (%)	AA (%)	κ (%)	Test (s)
0.001	95.88 ± 0.24	91.72 ± 0.31	95.18 ± 0.27	56 ± 2
0.01	$\mathbf{96.05 \pm 0.11}$	$\mathbf{92.10 \pm 0.18}$	$\mathbf{95.38 \pm 0.13}$	$\mathbf{55 \pm 2}$
0.10	95.97 ± 0.16	91.95 ± 0.24	95.29 ± 0.19	55 ± 3
1.00	96.01 ± 0.13	92.03 ± 0.21	95.33 ± 0.15	56 ± 2

V. PARAMETER OPTIMIZATION

Hyperparameter selection is important for CDF because the proposed framework combines a CD with multiple regularization objectives that jointly influence representation structure and classification performance. We adopt a Bayesian optimization strategy implemented using Optuna. The optimization process models validation accuracy as a surrogate objective and iteratively selects promising configurations based on expected improvement, enabling efficient exploration of the joint hyperparameter space while avoiding exhaustive grid search as shown in Table III.

The search space includes: learning rate (LR) $\{1e-5, 1e-4, 1e-3\}$, batch size $\{32, 64, 128\}$, embedding dimension $\{56, 128, 256\}$, orthogonality weight $\lambda_1 \in \{0.01, 0.1, 0.5, 1.0\}$, sparsity weight $\lambda_2 \in \{0.01, 0.1, 0.5, 1.0\}$, and contrastive weight $\lambda_3 \in \{0.01, 0.1, 0.5, 1.0\}$.

Unless otherwise stated, classification loss is assigned a weight of 1.0. Each trial trains CDF for 50 epochs and evaluates validation accuracy under identical data splits. All trials use the AdamW optimizer with fixed weight decay and identical LR scheduling to ensure fairness across configurations.

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TABLE V

QUANTITATIVE COMPARISON OF CDF WITH RECENT TRANSFORMER-BASED AND MAMBA-BASED HSI CLASSIFIERS UNDER THE SAME LOW-LABEL PROTOCOL (2% TRAINING, 2% VALIDATION, AND 96% TESTING). PERFORMANCE IS EVALUATED USING OA, AA, COHEN'S κ, INFERENCE TIME, AND THE NUMBER OF TRAINABLE PARAMETERS (PARAMS). BOLDFACE INDICATES THE BEST RESULT FOR EACH METRIC

Metric	Transformer-based Models				Mamba-based Models					CDFormer
	HViT	BiFormer	SSFormer	WaveFormer	WaveMamba	SSMamba	MFormer	MorpMamba	GraphMamba	
HC Dataset										
κ	91.1626	90.0576	90.9481	91.8101	85.4463	78.1826	85.2435	87.2331	92.9374	**95.3805**
OA	92.4482	91.5216	92.2553	92.9983	87.5916	81.4936	87.3825	89.1165	93.9719	**96.0546**
AA	83.1428	79.2418	86.7740	88.3590	71.6941	62.4808	76.1702	78.3453	85.3509	**92.1004**
Inference (s)	109	65	145	56	1922	258	262	2449	281	**55**
Params	813392	59754	2560528	888208	77584	**44176**	48464	65862	133905	301056
HH Dataset										
κ	89.1858	91.3602	93.9390	89.7752	81.5594	73.2130	87.0852	87.4309	94.4283	**95.6272**
OA	91.4534	93.1596	95.2045	91.9148	85.5363	78.9050	89.8245	90.1095	95.5935	**96.5406**
AA	77.2775	82.6329	88.8378	80.6076	62.5230	52.8081	71.6620	73.8562	88.0583	**91.9774**
Inference (s)	169	98	221	92	2709	384	396	438	423	**83**
Params	814934	60528	2562070	889750	78358	**44950**	49238	66636	134679	303702
TD Dataset										
κ	64.3987	88.4342	92.0796	90.2545	81.3439	54.2232	75.6024	74.8699	93.2487	**94.1490**
OA	69.2028	89.8097	93.0777	91.4570	83.8016	60.8320	79.0179	78.1578	94.0976	**94.8726**
AA	40.6790	68.1282	87.3970	78.9791	60.8519	33.5667	52.9607	55.0070	89.1557	**89.7633**
Inference (s)	144	97	146	84	1884	441	425	505	256	**73**
Params	813906	60012	2561042	888722	77842	**44434**	48722	66120	134163	301938
PG Dataset										
κ	56.9207	94.2306	90.2213	91.4493	85.4708	61.1624	76.4155	82.3480	91.4327	**94.9737**
OA	73.9701	96.1323	93.4768	94.2353	90.2509	75.6126	84.5625	88.1823	94.2878	**96.6288**
AA	21.1679	82.3080	80.7610	84.6378	70.0808	29.7600	50.8760	69.6580	80.9368	**86.9717**
Inference (s)	322	**211**	432	212	5185	1185	992	1060	582	256
Params	811850	58980	2558986	886666	76810	**43402**	47690	65088	133131	298410
QG Dataset										
κ	85.6042	85.9707	90.4918	87.5542	86.5816	38.9376	16.7375	80.8684	92.9904	**93.8855**
OA	89.1307	89.3692	92.8342	90.6144	89.9156	53.6150	36.0619	85.6345	94.7142	**95.3905**
AA	68.9710	86.6581	82.9808	84.5511	75.2695	37.0068	24.8338	75.7956	89.6632	**89.9022**
Inference (s)	337	423	470	**169**	4411	772	834	897	489	196
Params	810822	58464	2557958	885638	76294	**42886**	47174	64572	132615	296646

To assess the robustness of the selected regularization weights, we additionally performed a sensitivity analysis of the OR coefficient λ_1 over the range $\{0.001, 0.01, 0.1, 1.0\}$ on the HC dataset, as shown in Table IV. The resulting performance variations remain relatively small across the tested range, indicating that CDF is not overly sensitive to moderate changes in the OR strength.

The configurations in Table III illustrate that multiple hyperparameter combinations achieve competitive validation performance despite substantial differences in embedding dimension, LR, and regularization strength. The highest-performing configuration combines a larger embedding dimension with a conservative LR, which stabilizes optimization under the 2% supervision regime. The sensitivity analysis in Table IV further shows that performance remains stable across a broad range of orthogonality coefficients. Collectively, these observations indicate that CDF can maintain strong performance across a range of regularization settings rather than relying on a single narrowly tuned hyperparameter combination. Overall, these results suggest that balancing embedding capacity with structured regularization is important for achieving robust performance under limited-label HSI classification.

VI. QUALITATIVE AND QUANTITATIVE RESULTS AND DISCUSSION

We compare CDF with recent high-performing HSI classifiers that rely on global token mixing and sequence modeling. The baselines include Transformer-based methods (HViT [53], BiFormer (BiF) [54], SSFormer (SSF) [55], WaveFormer (WF) [56]) and state-space (Mamba-based) models (WaveMamba (WM) [57], SSMamba (SSM) [58], MFormer (MF) [59], MorpMamba (MM) [48], and GraphMamba (GM) [60]). All methods are evaluated under the same low-label protocol (2% training, 2% validation, 96% testing) and identical data partitions as described in Section IV. Hyperparameters are tuned using the procedure in Section V, and Table V reports the final results using OA, AA, and Cohen's κ.

A. Overall Performance (OA and κ)

As summarized in Table V, CDF achieves the strongest OA and κ on all evaluated datasets, indicating robust global correctness and agreement beyond chance under extreme label scarcity. On HC and HH, CDF establishes clear gains over both Transformer- and Mamba-based baselines, reaching 96.05%

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OA/95.38 κ on HC and 96.54% OA/95.63 κ on HH. The observed improvements remain consistent across datasets with different spatial resolutions, spectral dimensionalities, and scene characteristics. These results suggest that structured representation learning can provide complementary benefits to both attention-based and state-space sequence modeling architectures under limited-label supervision.

B. Class-Balanced Performance (AA) Under Scarce Supervision

AA is particularly informative under the 2% training setting because it penalizes failures on minority and difficult classes. CDF yields the highest AA on HC (92.10%), HH (91.98%), and TD (89.76%), demonstrating that the observed performance gains are not solely driven by dominant classes. Notably, several baselines maintain moderate-to-high OA while exhibiting substantially lower AA (e.g., HViT on TD and PG), suggesting less stable class-wise decision boundaries under limited supervision. The consistently stronger AA achieved by CDF is consistent with learning compact and discriminative representations through the combined use of concept projection, SP, and supervised contrastive learning. These results indicate improved class-wise consistency, particularly for challenging categories that are underrepresented in the training set.

C. Dataset-Dependent Difficulty and Stability

Performance differences are most pronounced on TD and PG, which represent challenging coastal and complex scenes with high interclass similarity. In TD, CDF maintains strong OA (94.87%) and AA (89.76%) while several competing methods experience substantial AA degradation. This observation is consistent with the hypothesis that structured representation learning can provide a more stable basis for discriminating visually similar classes when labeled supervision is scarce. On PG and QG, CDF also achieves the highest OA and κ, while AA differences across methods become more nuanced, implying that in some scenes the primary benefit may be improved global consistency and reduced noisy confusion rather than uniformly higher per-class accuracy.

D. Transformer-Based Models

Transformer baselines benefit from attention-driven long-range interactions, yet their AA can vary substantially in the low-label regime. For example, across several datasets, SSFormer and WaveFormer generally outperform BiFormer and HViT in AA, indicating that architectural choices in spectral–spatial token mixing and fusion remain important when supervision is limited. CDF consistently improves AA and κ beyond these attention-based baselines.

E. Mamba-Based Models

State-space models are attractive for long-sequence modeling, but Table V shows that several Mamba variants exhibit substantial drops in AA on difficult datasets such as TD and PG. This suggests that efficient sequence modeling alone may not guarantee reliable class-wise separation when only limited labeled data are available. GM is generally the strongest among the Mamba baselines in terms of OA and κ, which is consistent with the benefits of incorporating additional structural information.

TABLE VI

ABLATION STUDY OF THE PRINCIPAL COMPONENTS IN CDF. CD DENOTES CONCEPT DISENTANGLEMENT, OR DENOTES ORTHOGONALITY REGULARIZATION, SP DENOTES SPARSITY REGULARIZATION, AND CL DENOTES CONTRASTIVE LOSS. OA IS REPORTED FOR ALL DATASETS

CD	OR	SP	CL	HC	HH	PG	QG	TD
×	×	×	×	0.9297	0.9302	0.9250	0.9278	0.9389
✓	×	×	×	0.9577	0.9539	0.9705	0.9535	0.9534
✓	✓	×	×	0.9385	0.9599	0.9701	0.9605	0.9391
✓	×	✓	×	0.9444	0.9601	0.9700	0.9573	0.9510
✓	×	×	✓	0.9508	0.9654	0.9742	0.9521	0.9407
✓	✓	✓	×	0.9555	0.9674	0.9744	0.9541	0.9501
✓	✓	×	✓	0.9614	0.9694	0.9716	0.9599	0.9451
✓	×	✓	✓	0.9414	0.9623	0.9713	0.9573	0.9543
✓	✓	✓	✓	**0.9690**	**0.9777**	**0.9790**	**0.9673**	**0.9636**

F. Efficiency and Model Size

Beyond accuracy, Table V reports inference time and parameter count. CDF remains lightweight (approximately 3×10^5 parameters) while providing competitive inference efficiency compared with many sequence-based baselines. In particular, WM exhibits substantially larger inference time across datasets, whereas CDF achieves considerably faster testing while simultaneously improving classification performance. The reported runtimes correspond to inference under identical hardware and software settings without model compression techniques or deployment-specific acceleration. Importantly, the performance gains of CDF are achieved without increasing parameter count to the scale of the largest Transformer baselines, suggesting that the improvements are associated with representation structuring and optimization strategy rather than model scaling alone.

G. Qualitative Analysis of Classification Maps

Figs. 2–6 visualize the spatial distribution of predictions. CDF produces classification maps that are visually closer to the ground truth, with improved region connectivity, fewer isolated misclassified pixels, and sharper transitions at class boundaries. In contrast, several competing models exhibit fragmented regions, particularly in spectrally similar areas where minority classes are underrepresented. These qualitative observations are consistent with the quantitative improvements in OA, AA, and κ, and suggest that the proposed representation-learning strategy can contribute to more spatially coherent and class-consistent predictions under limited supervision.

VII. ABLATION STUDY

To quantify the contribution of the principal components in CDF, we perform a focused ablation study by progressively enabling the CD, OR, SP, and supervised CL. All experiments

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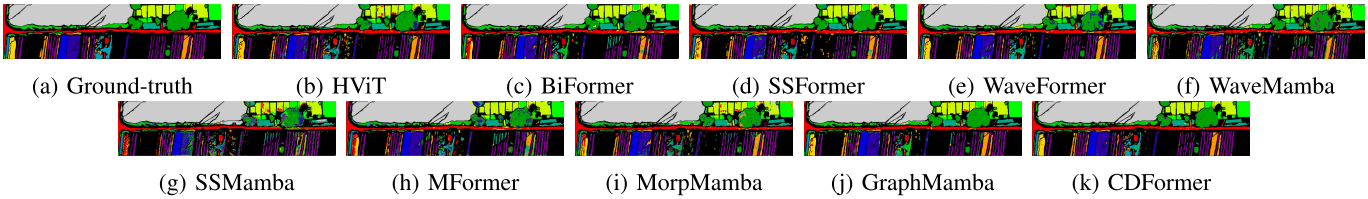


Fig. 2. Visual comparison of classification maps on the HC dataset. (a) Ground-truth map. (b)–(j) Predictions of competing transformer- and Mamba-based baselines. (k) Prediction of CDF.

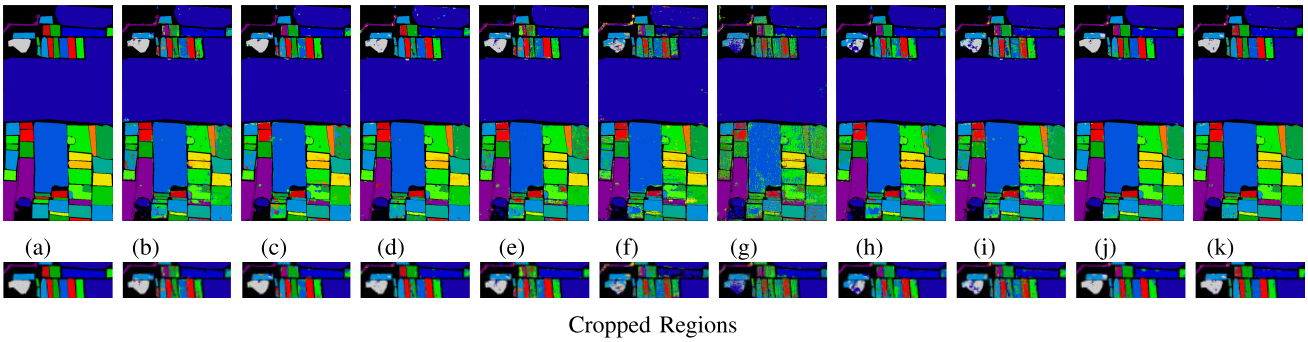


Fig. 3. Visual comparison of classification maps on the HH dataset. (a) GT. (b) HViT. (c) BiF. (d) SSF. (e) WF. (f) WM. (g) SSM. (h) MF. (i) MM. (j) GM. (k) CDF.

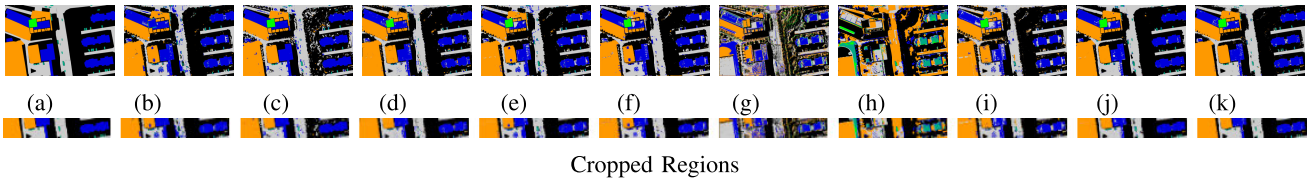


Fig. 4. Visual comparison of classification maps on the QG dataset. (a) GT. (b) HViT. (c) BiF. (d) SSF. (e) WF. (f) WM. (g) SSM. (h) MF. (i) MM. (j) GM. (k) CDF.

are conducted under the same low-label protocol using identical data partitions and training schedules. Table VI reports the resulting OA on the HC, HH, PG, QG, and TD datasets.

Comparing the baseline configuration (all components disabled) with the configuration that enables only CD reveals a substantial performance improvement across all datasets. For example, OA increases from 0.9297 to 0.9577 on HC and from 0.9302 to 0.9539 on HH. These results indicate that introducing an explicit CD provides a stronger and more structured representation than direct classification from unconstrained latent embeddings.

The addition of OR, SP, and CL further modifies the geometry of the learned representation space. While the effect of each regularizer varies across datasets, their inclusion generally improves performance when combined with the CD. OR reduces redundancy among concept directions, SP promotes selective activation patterns, and CL encourages compact same-class clusters and improved interclass separation.

Pairwise combinations of OR, SP, and CL typically outperform the baseline concept-bottleneck configuration, indicating that the three regularization mechanisms provide complementary benefits. Although no single regularizer consistently dominates across all datasets, the strongest performance is generally observed when multiple regularization terms are jointly employed.

The complete CDF model, which combines CD, OR, SP, and CL, achieves the highest OA on every evaluated dataset, reaching 0.9690 (HC), 0.9777 (HH), 0.9790 (PG), 0.9673 (QG), and 0.9636 (TD). These results suggest that the proposed components work synergistically and that the final performance gains arise from their combined optimization rather than from any single module in isolation.

VIII. Structural Transparency and Concept Analysis

Concept analysis is useful for understanding how information is organized within the learned representation space and how different concept dimensions contribute to the prediction process. Because CDF performs classification through an explicit class-concept mapping, the learned concept activations can be analyzed directly through attribution maps and class-concept dependencies. The following experiments provide empirical insight into the structure of the learned concept space and the corresponding decision pathway.

A. Class-Concept Dependency

To examine how concepts contribute to class predictions, we analyze the classifier matrix $\mathbf{W}_{\text{cls}} \in \mathbb{R}^{K \times C_l}$ (here $K = 6$ and $C_l = 22$ for HH). For a sample with pooled concept vector $\bar{\mathbf{r}} \in \mathbb{R}^K$, the logit of class c is $\ell_c = \mathbf{W}_{\text{cls}}[:, c]^\top \bar{\mathbf{r}}$, so $\mathbf{W}_{\text{cls}}[k, c]$ quantifies the signed contribution of concept k to class c. To quantify concept specificity, we define a winner-takes-all class selectivity score. Let r_{ik} denote the activation of concept k for

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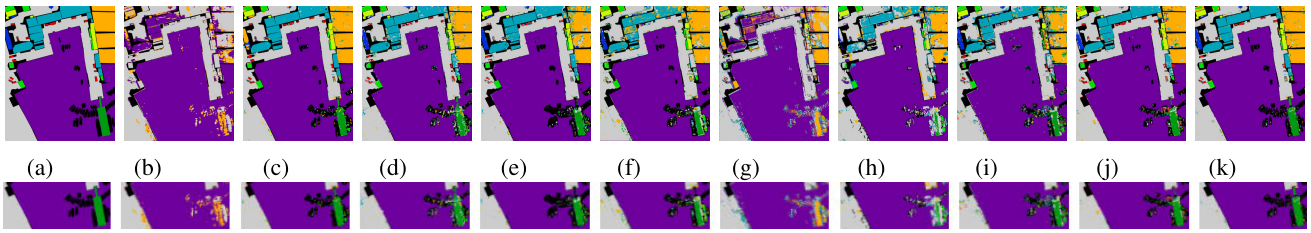


Fig. 5. Visual comparison of classification maps on the PG dataset. (a) GT. (b) HViT. (c) BiF. (d) SSF. (e) WF. (f) WM. (g) SSM. (h) MF. (i) MM. (j) GM. (k) CDF.

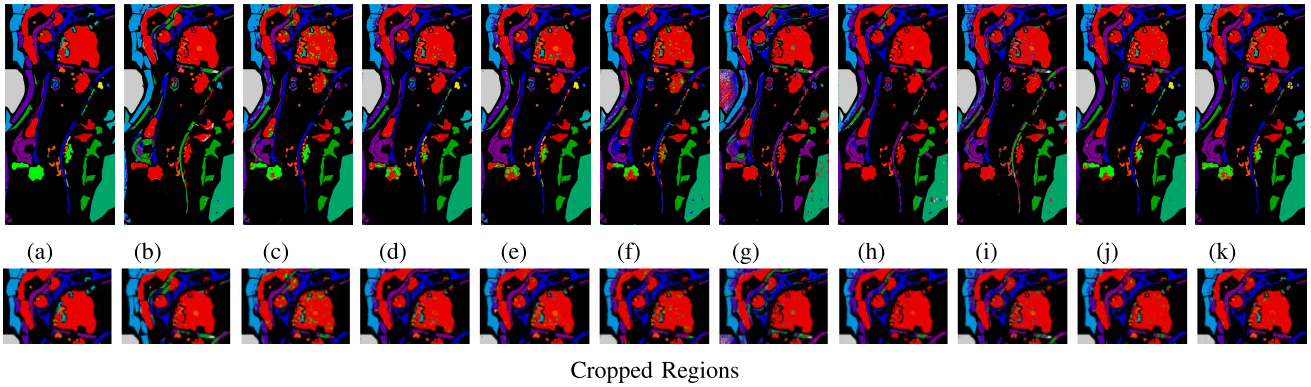


Fig. 6. Visual comparison of classification maps on the TD dataset. (a) GT. (b) HViT. (c) BiF. (d) SSF. (e) WF. (f) WM. (g) SSM. (h) MF. (i) MM. (j) GM. (k) CDF.

pixel (patch center) i. Define the top-1 concept assignment $\hat{k}_i = \arg\max_k r_{ik}$. The selectivity of concept k is

$$\text{Selectivity}\,(k) = \max_c \left(\frac{1}{N_c} \sum_{i:y_i=c} \mathbb{1}\left[\hat{k}_i = k\right] \right) \tag{26}$$

where N_c is the number of samples in class c. A value near 1 indicates that concept k is predominantly the top-activated concept for a single class, whereas lower values indicate broader reuse across classes.

B. Robustness of Concept Representations

The concept pathway also enables stability analysis under spectral perturbations. For the linear concept projection stage $\mathbf{z} = \mathbf{W}_c^\top \mathbf{h}$ (with feature $\mathbf{h}$ and projection $\mathbf{W}_c$), a perturbation $\Delta\mathbf{h}$ satisfies $\|\Delta\mathbf{z}\|_2 \leq \|\mathbf{W}_c\|_2 \|\Delta\mathbf{h}\|_2$. When combined with a Lipschitz encoder and activation function (Theorem 2), this yields an end-to-end Lipschitz bound in concept space, suggesting bounded sensitivity of concept activations under bounded input noise.

1) Semantic Association Analysis: To further examine whether the learned concept representations exhibit meaningful associations with scene semantics, we perform a correlation-based analysis on the HC dataset. Following common remote-sensing categories, classes are grouped into broader semantic categories, including crop, vegetation, water, built-up, transport infrastructure, bare soil, and other. For each semantic group, we compute the correlation between the corresponding binary group indicator and each learned concept dimension. Table VII reports the three concept dimensions exhibiting the strongest absolute correlations for each semantic category.

TABLE VII
SEMANTIC ASSOCIATION ANALYSIS ON THE HC DATASET. FOR EACH SEMANTIC GROUP, THE THREE CONCEPT DIMENSIONS WITH THE STRONGEST ABSOLUTE CORRELATION ARE REPORTED. POSITIVE AND NEGATIVE VALUES INDICATE THE DIRECTION OF ASSOCIATION BETWEEN THE SEMANTIC GROUP INDICATOR AND THE CORRESPONDING LEARNED CONCEPT DIMENSION

Semantic Group	Most Associated Concept Dimensions
Bare Soil	C13 (+0.475), C16 (-0.342), C15 (-0.304)
Built-up	C5 (-0.554), C10 (+0.524), C11 (+0.442)
Crop	C13 (+0.469), C12 (-0.379), C11 (-0.283)
Other	C12 (+0.519), C4 (+0.448), C14 (+0.356)
Transport Infrastructure	C14 (+0.499), C7 (-0.447), C16 (-0.337)
Vegetation	C1 (-0.404), C16 (+0.360), C10 (-0.328)
Water	C16 (+0.479), C8 (-0.365), C5 (+0.292)

The results indicate that several concept dimensions exhibit consistent associations with semantically related groups. For example, Concept 13 shows the strongest positive correlation with both crop (+0.469) and bare-soil (+0.475) categories, Concept 10 is strongly associated with built-up regions (+0.524), Concept 14 is associated with transport infrastructure (+0.499), and Concept 16 exhibits the strongest positive correlation with water-related regions (+0.479). These observations suggest that the learned concept space captures meaningful semantic structure despite being trained without concept annotations. Importantly, these associations should be interpreted as emergent properties of the learned representation geometry rather than evidence of predefined semantic concepts. The learned concept dimensions remain

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latent coordinates optimized for classification and representation organization, and any semantic alignment is dataset dependent rather than explicitly enforced during training.

IX. Conclusion and Future Work

This article presented CDF, a Transformer-based framework for HSI classification that integrates an explicit CD to promote structured representation learning and analyzable decision pathways. Instead of performing classification directly from high-dimensional latent embeddings, CDF projects contextualized spectral–spatial features into a lower-dimensional concept space and predicts classes through a linear class-concept mapping. This design enables an explicit decomposition of the decision function while preserving the modeling strengths of Transformer architectures.

To improve representation quality under limited supervision, CDF combines concept projection with OR, SP, and supervised CL within a unified optimization framework. The resulting concept space provides a structured intermediate representation that supports decision decomposition, representation analysis, and stable learning under low-label conditions.

Extensive experiments on five benchmark HSI datasets demonstrated that CDF consistently achieves strong performance compared with recent Transformer- and state-space-based baselines under a strict low-label protocol. Additional concept attribution, class-concept dependency, and semantic association analyses indicate that the learned concept space captures meaningful representation structure while remaining fully annotation-free.

Several promising research directions remain. Future work will investigate nonlinear and manifold-aware concept mappings, formal measures of concept stability and selectivity, and stronger semantic validation through expert-guided evaluation. Extending concept-based structured representation learning to hyperspectral time-series analysis and multimodal remote sensing settings, including LiDAR and SAR fusion, also represents a promising direction for improving robustness and decision transparency in complex Earth observation applications.

References

[1] D. Hong et al., "Hyperspectral imaging," *Nature Rev. Methods Primers*, vol. 6, no. 1, p. 19, 2026.

[2] M. Ahmad, M. Mazzara, S. Distefano, A. Mehmood Khan, M. Hassaan FarooqButt, and D. Hong, "PolicyMamba: Localized policy attention with state space model for land cover classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17814–17825, Oct. 2025.

[3] M. Moroni, E. Lupo, E. Marra, and A. Cenedese, "Hyperspectral image analysis in environmental monitoring: Setup of a new tunable filter platform," *Proc. Environ. Sci.*, vol. 19, pp. 885–894, Jul. 2013. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S187802961300368X

[4] B. G. Ram, P. Oduor, C. Igathinathane, K. Howatt, and X. Sun, "A systematic review of hyperspectral imaging in precision agriculture: Analysis of its current state and future prospects," *Comput. Electron. Agricult.*, vol. 222, Jul. 2024, Art. no. 109037. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0168169924004289

[5] C. Li et al., "UrbanSAM: Learning invariance-inspired adapters for segment anything models in urban construction," 2025, *arXiv:2502.15199*.

[6] A. Guha, "15-mineral exploration using hyperspectral data," in *Hyperspectral Remote Sensing (Earth Observation)*. Amsterdam, The Netherlands: Elsevier, 2020, pp. 293–318. [Online]. Available: https://www.sciencedirect.com/science/article/pii/B9780081028940000127

[7] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[8] J. Xie et al., "Is human-like decision making explainable? Towards an explainable artificial intelligence for autonomous vehicles," *Transp. Res. Interdiscipl. Perspect.*, vol. 29, Jan. 2025, Art. no. 101278. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590198224002641

[9] B. G. Ram, K. Howatt, J. Mettler, and X. Sun, "Addressing computation resource exhaustion associated with deep learning training of three-dimensional hyperspectral images using multiclass weed classification," *Artif. Intell. Agricult.*, vol. 15, no. 2, pp. 131–146, Jun. 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2589721725000248

[10] M. Mulimani and A. Mesaros, "A closer look at class-incremental learning for multi-label audio classification," *IEEE Trans. Audio, Speech Lang. Process.*, vol. 33, pp. 1293–1306, 2025.

[11] D. Li et al., "CSTFNet: A CNN and dual Swin-transformer fusion network for remote sensing hyperspectral data fusion and classification of coastal areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5853–5865, 2025.

[12] M. Ahmad, H. Luqman, M. Mazzara, and S. L. Ullo, "Saliency-modulated group dynamic attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 21230–21244, 2026.

[13] M. Ahmad and M. Mazzara, "Iformer: Irregular spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5521012.

[14] M. Ahmad, M. Mazzara, S. Distefano, and A. M. Khan, "Byte latent mamba with state space and knowledge distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5531815.

[15] M. Ahmad et al., "GraphMamba: Graph tokenization mamba for hyperspectral image classification," *IEEE Trans. Emerg. Topics Comput.*, vol. 13, no. 4, pp. 1–12, Oct. 2025.

[16] V. Noferesti and H. Mirzahossein, "Leveraging machine learning to predict residential location choice: A comparative analysis," *Results Eng.*, vol. 25, Mar. 2025, Art. no. 104214. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590123025003007

[17] B. Chander, C. John, L. Warrier, and K. Gopalakrishnan, "Toward trustworthy artificial intelligence (TAI) in the context of explainability and robustness," *ACM Comput. Surveys*, vol. 57, no. 6, pp. 1–49, Jun. 2025, doi: 10.1145/3675392.

[18] X. Li, C. Li, P. Ghamisi, D. Hong, J. A. Benediktsson, and J. Chanussot, "FlexiMo: A flexible remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5606516.

[19] M. Yang, W. Quan, and M. Wei, "Perceive, act and correct: Confidence is not enough for hyperspectral classification," in *Proc. AAAI Conf. Artif. Intell.*, 2026, vol. 40, no. 14, pp. 11631–11639, doi: 10.1609/aaai.v40i14.38147.

[20] W. Quan, M. Wei, and J. Tang, "Equal is not always fair: A new perspective on hyperspectral representation non-uniformity," 2025, *arXiv:2505.11267*.

[21] W. Quan, W. Zhao, W. Wang, H. Xie, F. Lee Wang, and M. Wei, "Lost in UNet: Improving infrared small target detection by underappreciated local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5000115.

[22] L. Huang et al., "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Trans. Inf. Syst.*, vol. 43, no. 2, pp. 1–55, Mar. 2025, doi: 10.1145/3703155.

[23] Y. Sun, Y. Ji, H. Zhu, F. Zhuang, Q. He, and H. Xiong, "Market-aware long-term job skill recommendation with explainable deep reinforcement learning," *ACM Trans. Inf. Syst.*, vol. 43, no. 2, pp. 1–35, Mar. 2025, doi: 10.1145/3704998.

[24] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6883–6896, Apr. 2025.

[25] Y. Zheng, Y. Chen, B. Qian, X. Shi, Y. Shu, and J. Chen, "A review on edge large language models: Design, execution, and applications," *ACM Comput. Surveys*, vol. 57, no. 8, pp. 1–35, Aug. 2025, doi: 10.1145/3719664.

[26] X. Liu et al., "Advancing interpretable cardiac disease diagnosis via a transformer-convolutional hybrid network on electrocardiograms," *Eng. Appl. Artif. Intell.*, vol. 152, Jul. 2025, Art. no. 110675. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095219762500675X

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[27] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
[28] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[29] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.
[30] H. Bi, H. Chang, X. Wang, and D. Hong, "Cross-attention-driven adaptive graph relational network for multilabel remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.
[31] H. Pei, T. Owari, S. Tsuyuki, T. Hiroshima, and D. Hong, "Combined gated graph convolution neural networks with multi-modal geospatial data for forest type classification," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 136, Feb. 2025, Art. no. 104372.
[32] G. De Lucia, M. Lapegna, and D. Romano, "Towards explainable ai for hyperspectral image classification in edge computing environments," *Comput. Electr. Eng.*, vol. 103, Sep. 2022, Art. no. 108381. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0045790622005985
[33] S. O. Atik and M. E. Atik, "Optimal band selection using explainable artificial intelligence for machine learning-based hyperspectral image classification," *J. Appl. Remote Sens.*, vol. 18, no. 4, Jul. 2024, Art. no. 042604, doi: 10.1117/1.jrs.18.042604.
[34] A. Temenos, N. Temenos, M. Kaselimi, A. Doulamis, and N. Doulamis, "Interpretable deep learning framework for land use and land cover classification in remote sensing using SHAP," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[35] S. Ebrahimi and S. Kumar, "What helps to detect what? Explainable AI and multisensor fusion for semantic segmentation of simultaneous crop and land cover land use delineation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5423–5444, 2025.
[36] J. Deng, D. Hong, C. Li, and N. Yokoya, "Joint super-resolution and segmentation for 1-m impervious surface area mapping in China's Yangtze river economic belt," *GIScience Remote Sens.*, vol. 63, no. 1, Dec. 2026, Art. no. 2610548.
[37] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.
[38] I. Kakogeorgiou and K. Karantzalos, "Evaluating explainable artificial intelligence methods for multi-label deep learning classification tasks in remote sensing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102520. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0303243421002270
[39] Y. Shi et al., "A biologically interpretable two-stage deep neural network (BIT-DNN) for vegetation recognition from hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4401320.
[40] L. Huang, Y. Chen, and X. He, "Foundation model-based spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5529825.
[41] W. Kong, B. Liu, X. Bi, C. Yu, X. Li, and Y. Chen, "HyperSL: A spectral foundation model for hyperspectral image interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–19, 2025.
[42] J. Fu et al., "HyperR3SNet: Leveraging hyperbolic space and vision foundation models for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5620016.
[43] Y. Zhu and L. L. Xu, "Language-informed hyperspectral image synthesis for imbalanced-small sample classification via semi-supervised conditional diffusion model," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1–21, 2026.
[44] W. Liu, S. Prasad, and M. Crawford, "Advancing efficient vision foundation models for analysis of multispectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 10825–10841, 2026.
[45] J. Deng, T. Chen, C. Li, and D. Hong, "Foundation models for phenotyping segmentation of wheat stripe rust resistance from UAV hyperspectral images," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV) Workshops*, Mar. 2026, pp. 849–858.
[46] Y. Xia et al., "SpectralZero: Text-driven spectral–spatial alignment for zero-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5507016.
[47] D. Hong, C. Li, X. Li, G. Camps-Valls, and J. Chanussot, "Foundation models in remote sensing: Evolving from unimodality to multimodality," *IEEE Geosci. Remote Sens. Mag.*, vol. 14, no. 2, pp. 10–35, Apr. 2026.
[48] M. Ahmad et al., "Spatial–spectral morphological mamba for hyperspectral image classification," *Neurocomputing*, vol. 636, Mar. 2025, Art. no. 129995. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231225006678
[49] X. Li, C. Li, G. Vivone, and D. Hong, "SeaMo: A season-aware multimodal foundation model for remote sensing," *Inf. Fusion*, vol. 125, Jan. 2025, Art. no. 103334.
[50] M. Talagrand, "Concentration of measure and isoperimetric inequalities in product spaces," *Publications Mathématiques de l'Institut des Hautes Études Scientifiques*, vol. 81, pp. 73–205, Dec. 1995, doi: 10.1007/bf02699376.
[51] M. O. Searcoid, *Metric Spaces*. Cham, Switzerland: Springer, 2006.
[52] T. Donchev and E. Farkhi, "Stability and Euler approximation of one-sided Lipschitz differential inclusions," *SIAM J. Control Optim.*, vol. 36, no. 2, pp. 780–796, Mar. 1998, doi: 10.1137/s0363012995293694.
[53] Q. Yu, W. Wei, D. Li, Z. Pan, C. Li, and D. Hong, "HyperSINet: A synergetic interaction network combined with convolution and transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508118.
[54] Y. Wang, L. Yang, T. Xinzhang Wu, K. Tang, and W. Zha, "Spatial–spectral fusion biformer: A novel dynamic routing approach for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.
[55] Q. Ran et al., "Deep transformer and few-shot learning for hyperspectral image classification," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1323–1336, 2023.
[56] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[57] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "WaveMamba: Spatial–spectral wavelet mamba for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[58] Z. Pan, C. Li, A. Plaza, J. Chanussot, and D. Hong, "Hyperspectral image classification with mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602814, doi: 10.1109/TGRS.2024.3521411.
[59] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "HSI-MFormer: Integrating mamba and transformer experts for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621916.
[60] A. Yang, M. Li, Y. Ding, L. Fang, Y. Cai, and Y. He, "GraphMamba: An efficient graph structure learning vision mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

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